Frontend

Src

Components

Ui:

badge.tsx:

```
import * as React from "react"

import { cva, type VariantProps } from "class-variance-authority"

import { Slot } from "radix-ui"


import { cn } from "@/lib/utils"


const badgeVariants = cva(

  "group/badge inline-flex h-5 w-fit shrink-0 items-center justify-center gap-1 overflow-hidden rounded-4xl border border-transparent px-2 py-0.5 text-xs font-medium whitespace-nowrap transition-all focus-visible:border-ring focus-visible:ring-[3px] focus-visible:ring-ring/50 has-data-[icon=inline-end]:pr-1.5 has-data-[icon=inline-start]:pl-1.5 aria-invalid:border-destructive aria-invalid:ring-destructive/20 dark:aria-invalid:ring-destructive/40 [&>svg]:pointer-events-none [&>svg]:size-3!",

  {

    variants: {

      variant: {

        default: "bg-primary text-primary-foreground [a]:hover:bg-primary/80",

        secondary:

          "bg-secondary text-secondary-foreground [a]:hover:bg-secondary/80",

        destructive:

          "bg-destructive/10 text-destructive focus-visible:ring-destructive/20 dark:bg-destructive/20 dark:focus-visible:ring-destructive/40 [a]:hover:bg-destructive/20",

        outline:

          "border-border text-foreground [a]:hover:bg-muted [a]:hover:text-muted-foreground",
```

```
      ghost:
        "hover:bg-muted hover:text-muted-foreground dark:hover:bg-muted/50",
      link: "text-primary underline-offset-4 hover:underline",
    },
  },
  defaultVariants: {
    variant: "default",
  },
}
)

function Badge({
  className,
  variant = "default",
  asChild = false,
  ...props
}: React.ComponentProps<"span"> &
  VariantProps<typeof badgeVariants> & { asChild?: boolean }) {
  const Comp = asChild ? Slot.Root : "span"

  return (
    <Comp
      data-slot="badge"
      data-variant={variant}
      className={cn(badgeVariants({ variant }), className)}
      {...props}
    />
  )
```

```
}

export { Badge, badgeVariants }
```

button.tsx:

```
import * as React from "react"

import { cva, type VariantProps } from "class-variance-authority"

import { Slot } from "radix-ui"

import { cn } from "@/lib/utils"

const buttonVariants = cva(

  "group/button inline-flex shrink-0 items-center justify-center rounded-lg border border-transparent bg-clip-padding text-sm font-medium whitespace-nowrap transition-all outline-none select-none focus-visible:border-ring focus-visible:ring-3 focus-visible:ring-ring/50 disabled:pointer-events-none disabled:opacity-50 aria-invalid:border-destructive aria-invalid:ring-3 aria-invalid:ring-destructive/20 dark:aria-invalid:border-destructive/50 dark:aria-invalid:ring-destructive/40 [&_svg]:pointer-events-none [&_svg]:shrink-0 [&_svg:not([class*='size-'])]:size-4",

  {

    variants: {

      variant: {

        default: "bg-primary text-primary-foreground [a]:hover:bg-primary/80",

        outline:

          "border-border bg-background hover:bg-muted hover:text-foreground aria-expanded:bg-muted aria-expanded:text-foreground dark:border-input dark:bg-input/30 dark:hover:bg-input/50",

        secondary:

          "bg-secondary text-secondary-foreground hover:bg-secondary/80 aria-expanded:bg-secondary aria-expanded:text-secondary-foreground",

        ghost:
```

```
      "hover:bg-muted hover:text-foreground aria-expanded:bg-muted aria-expanded:text-foreground dark:hover:bg-muted/50",
      destructive:
      "bg-destructive/10 text-destructive hover:bg-destructive/20 focus-visible:border-destructive/40 focus-visible:ring-destructive/20 dark:bg-destructive/20 dark:hover:bg-destructive/30 dark:focus-visible:ring-destructive/40",
      link: "text-primary underline-offset-4 hover:underline",
    },
    size: {
      default:
      "h-8 gap-1.5 px-2.5 has-data-[icon=inline-end]:pr-2 has-data-[icon=inline-start]:pl-2",
      xs: "h-6 gap-1 rounded-[min(var(--radius-md),10px)] px-2 text-xs in-data-[slot=button-group]:rounded-lg has-data-[icon=inline-end]:pr-1.5 has-data-[icon=inline-start]:pl-1.5 [&_svg:not([class*='size-'])]:size-3",
      sm: "h-7 gap-1 rounded-[min(var(--radius-md),12px)] px-2.5 text-[0.8rem] in-data-[slot=button-group]:rounded-lg has-data-[icon=inline-end]:pr-1.5 has-data-[icon=inline-start]:pl-1.5 [&_svg:not([class*='size-'])]:size-3.5",
      lg: "h-9 gap-1.5 px-2.5 has-data-[icon=inline-end]:pr-3 has-data-[icon=inline-start]:pl-3",
      icon: "size-8",
      "icon-xs":
      "size-6 rounded-[min(var(--radius-md),10px)] in-data-[slot=button-group]:rounded-lg [&_svg:not([class*='size-'])]:size-3",
      "icon-sm":
      "size-7 rounded-[min(var(--radius-md),12px)] in-data-[slot=button-group]:rounded-lg",
      "icon-lg": "size-9",
    },
  },
  defaultVariants: {
```

```
    variant: "default",
    size: "default",
  },
}
)

function Button({
  className,
  variant = "default",
  size = "default",
  asChild = false,
  ...props
}: React.ComponentProps<"button"> &
  VariantProps<typeof buttonVariants> & {
    asChild?: boolean
  }) {
  const Comp = asChild ? Slot.Root : "button"

  return (
    <Comp
      data-slot="button"
      data-variant={variant}
      data-size={size}
      className={cn(buttonVariants({ variant, size, className }))}
      {...props}
    />
  )
}
```

```
export { Button, buttonVariants }
```

card.tsx:

```
import * as React from "react"

import { cn } from "@/lib/utils"

function Card({
  className,
  size = "default",
  ...props
}: React.ComponentProps<"div"> & { size?: "default" | "sm" }) {
  return (
    <div
      data-slot="card"
      data-size={size}
      className={cn(
        "group/card flex flex-col gap-4 overflow-hidden rounded-xl bg-card py-4 text-sm text-card-foreground ring-1 ring-foreground/10 has-data-[slot=card-footer]:pb-0 has-[>img:first-child]:pt-0 data-[size=sm]:gap-3 data-[size=sm]:py-3 data-[size=sm]:has-data-[slot=card-footer]:pb-0 *:[img:first-child]:rounded-t-xl *:[img:last-child]:rounded-b-xl",
        className
      )}
      {...props}
    />
  )
}
```

```
function CardHeader({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="card-header"
      className={cn(
        "group/card-header @container/card-header grid auto-rows-min items-start gap-1 rounded-t-xl px-4 group-data-[size=sm]/card:px-3 has-data-[slot=card-action]:grid-cols-[1fr_auto] has-data-[slot=card-description]:grid-rows-[auto_auto] [.border-b]:pb-4 group-data-[size=sm]/card:[.border-b]:pb-3",
        className
      )}
      {...props}
    />
  )
}

function CardTitle({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="card-title"
      className={cn(
        "text-base leading-snug font-medium group-data-[size=sm]/card:text-sm",
        className
      )}
      {...props}
    />
  )
}
```

```
function CardDescription({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="card-description"
      className={cn("text-sm text-muted-foreground", className)}
      {...props}
    />
  )
}

function CardAction({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="card-action"
      className={cn(
        "col-start-2 row-span-2 row-start-1 self-start justify-self-end",
        className
      )}
      {...props}
    />
  )
}

function CardContent({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="card-content"
```

```
    className={cn("px-4 group-data-[size=sm]/card:px-3", className)}
    {...props}
   />
  )
}

function CardFooter({ className, ...props }: React.ComponentProps<"div">) {
  return (
   <div
    data-slot="card-footer"
    className={cn(
     "flex items-center rounded-b-xl border-t bg-muted/50 p-4 group-data-[size=sm]/card:p-3",
     className
    )}
    {...props}
   />
  )
}

export {
  Card,
  CardHeader,
  CardFooter,
  CardTitle,
  CardAction,
  CardDescription,
  CardContent,
```

```
}
```

Input.tsx:

```
import * as React from "react"

import { cn } from "@/lib/utils"

function Input({ className, type, ...props }: React.ComponentProps<"input">) {
  return (
    <input
      type={type}
      data-slot="input"
      className={cn(
        "h-8 w-full min-w-0 rounded-lg border border-input bg-transparent px-2.5 py-1 text-base transition-colors outline-none file:inline-flex file:h-6 file:border-0 file:bg-transparent file:text-sm file:font-medium file:text-foreground placeholder:text-muted-foreground focus-visible:border-ring focus-visible:ring-3 focus-visible:ring-ring/50 disabled:pointer-events-none disabled:cursor-not-allowed disabled:bg-input/50 disabled:opacity-50 aria-invalid:border-destructive aria-invalid:ring-3 aria-invalid:ring-destructive/20 md:text-sm dark:bg-input/30 dark:disabled:bg-input/80 dark:aria-invalid:border-destructive/50 dark:aria-invalid:ring-destructive/40",
        className
      )}
      {...props}
    />
  )
}

export { Input }
```

label.tsx:

```
import * as React from "react"

import { Label as LabelPrimitive } from "radix-ui"


import { cn } from "@/lib/utils"


function Label({

  className,

  ...props

}: React.ComponentProps<typeof LabelPrimitive.Root>) {

  return (

    <LabelPrimitive.Root

      data-slot="label"

      className={cn(

        "flex items-center gap-2 text-sm leading-none font-medium select-none group-data-[disabled=true]:pointer-events-none group-data-[disabled=true]:opacity-50 peer-disabled:cursor-not-allowed peer-disabled:opacity-50",

        className

      )}

      {...props}

    />

  )

}


export { Label }
```

progress.tsx:

```
import * as React from "react"

import { Progress as ProgressPrimitive } from "radix-ui"
```

```
import { cn } from "@/lib/utils"

function Progress({
  className,
  value,
  ...props
}: React.ComponentProps<typeof ProgressPrimitive.Root>) {
  return (
    <ProgressPrimitive.Root
      data-slot="progress"
      className={cn(
        "relative flex h-1 w-full items-center overflow-x-hidden rounded-full bg-muted",
        className
      )}
      {...props}
    >
      <ProgressPrimitive.Indicator
        data-slot="progress-indicator"
        className="size-full flex-1 bg-primary transition-all"
        style={{ transform: `translateX(-${100 - (value || 0)}%)` }}
      />
    </ProgressPrimitive.Root>
  )
}

export { Progress }
```

select.tsx:

```
"use client"

import * as React from "react"
import { Select as SelectPrimitive } from "radix-ui"

import { cn } from "@/lib/utils"
import { ChevronDownIcon, CheckIcon, ChevronUpIcon } from "lucide-react"

function Select({
  ...props
}: React.ComponentProps<typeof SelectPrimitive.Root>) {
  return <SelectPrimitive.Root data-slot="select" {...props} />
}

function SelectGroup({
  className,
  ...props
}: React.ComponentProps<typeof SelectPrimitive.Group>) {
  return (
    <SelectPrimitive.Group
      data-slot="select-group"
      className={cn("scroll-my-1 p-1", className)}
      {...props}
    />
  )
}
```

```
function SelectValue({
  ...props
}: React.ComponentProps<typeof SelectPrimitive.Value>) {
  return <SelectPrimitive.Value data-slot="select-value" {...props} />
}

function SelectTrigger({
  className,
  size = "default",
  children,
  ...props
}: React.ComponentProps<typeof SelectPrimitive.Trigger> & {
  size?: "sm" | "default"
}) {
  return (
    <SelectPrimitive.Trigger
      data-slot="select-trigger"
      data-size={size}
      className={cn(
        "flex w-fit items-center justify-between gap-1.5 rounded-lg border border-input bg-transparent py-2 pr-2 pl-2.5 text-sm whitespace-nowrap transition-colors outline-none select-none focus-visible:border-ring focus-visible:ring-3 focus-visible:ring-ring/50 disabled:cursor-not-allowed disabled:opacity-50 aria-invalid:border-destructive aria-invalid:ring-3 aria-invalid:ring-destructive/20 data-placeholder:text-muted-foreground data-[size=default]:h-8 data-[size=sm]:h-7 data-[size=sm]:rounded-[min(var(--radius-md),10px)] *:data-[slot=select-value]:line-clamp-1 *:data-[slot=select-value]:flex *:data-[slot=select-value]:items-center *:data-[slot=select-value]:gap-1.5 dark:bg-input/30 dark:hover:bg-input/50 dark:aria-invalid:border-destructive/50 dark:aria-invalid:ring-destructive/40 [&_svg]:pointer-events-none [&_svg]:shrink-0 [&_svg:not([class*='size-'])]:size-4",
        className
```

```
    )}
    {...props}
   >
    {children}
    <SelectPrimitive.Icon asChild>
      <ChevronDownIcon className="pointer-events-none size-4 text-muted-foreground" />
    </SelectPrimitive.Icon>
   </SelectPrimitive.Trigger>
  )
}

function SelectContent({
  className,
  children,
  position = "item-aligned",
  align = "center",
  ...props
}: React.ComponentProps<typeof SelectPrimitive.Content>) {
  return (
   <SelectPrimitive.Portal>
    <SelectPrimitive.Content
      data-slot="select-content"
      data-align-trigger={position === "item-aligned"}
      className={cn("relative z-50 max-h-(--radix-select-content-available-height) min-w-36 origin-(--radix-select-content-transform-origin) overflow-x-hidden overflow-y-auto rounded-lg bg-popover text-popover-foreground shadow-md ring-1 ring-foreground/10 duration-100 data-[align-trigger=true]:animate-none data-[side=bottom]:slide-in-from-top-2 data-[side=left]:slide-in-from-right-2 data-[side=right]:slide-in-from-left-2 data-[side=top]:slide-in-from-bottom-2 data-open:animate-in data-open:fade-in-0 data-
```

```
open:zoom-in-95 data-closed:animate-out data-closed:fade-out-0 data-closed:zoom-out-95", position ==="popper"&&"data-[side=bottom]:translate-y-1 data-[side=left]:-translate-x-1 data-[side=right]:translate-x-1 data-[side=top]:-translate-y-1", className )}
      position={position}
      align={align}
      {...props}
     >
      <SelectScrollUpButton />
      <SelectPrimitive.Viewport
        data-position={position}
        className={cn(
          "data-[position=popper]:h-(--radix-select-trigger-height) data-[position=popper]:w-full data-[position=popper]:min-w-(--radix-select-trigger-width)",
          position === "popper" && ""
        )}
      >
        {children}
      </SelectPrimitive.Viewport>
      <SelectScrollDownButton />
    </SelectPrimitive.Content>
   </SelectPrimitive.Portal>
  )
}

function SelectLabel({
  className,
  ...props
}: React.ComponentProps<typeof SelectPrimitive.Label>) {
```

```
  return (
    <SelectPrimitive.Label
      data-slot="select-label"
      className={cn("px-1.5 py-1 text-xs text-muted-foreground", className)}
      {...props}
    />
  )
}

function SelectItem({
  className,
  children,
  ...props
}: React.ComponentProps<typeof SelectPrimitive.Item>) {
  return (
    <SelectPrimitive.Item
      data-slot="select-item"
      className={cn(
        "relative flex w-full cursor-default items-center gap-1.5 rounded-md py-1 pr-8 pl-1.5 text-sm outline-hidden select-none focus:bg-accent focus:text-accent-foreground not-data-[variant=destructive]:focus:**:text-accent-foreground data-disabled:pointer-events-none data-disabled:opacity-50 [&_svg]:pointer-events-none [&_svg]:shrink-0 [&_svg:not([class*='size-'])]:size-4 *:[span]:last:flex *:[span]:last:items-center *:[span]:last:gap-2",
        className
      )}
      {...props}
    >
      <span className="pointer-events-none absolute right-2 flex size-4 items-center justify-center">
```

```
      <SelectPrimitive.ItemIndicator>
        <CheckIcon className="pointer-events-none" />
      </SelectPrimitive.ItemIndicator>
    </span>
    <SelectPrimitive.ItemText>{children}</SelectPrimitive.ItemText>
  </SelectPrimitive.Item>
 )
}

function SelectSeparator({
  className,
  ...props
}: React.ComponentProps<typeof SelectPrimitive.Separator>) {
  return (
    <SelectPrimitive.Separator
      data-slot="select-separator"
      className={cn("pointer-events-none -mx-1 my-1 h-px bg-border", className)}
      {...props}
    />
  )
}

function SelectScrollUpButton({
  className,
  ...props
}: React.ComponentProps<typeof SelectPrimitive.ScrollUpButton>) {
  return (
    <SelectPrimitive.ScrollUpButton
```

```
    data-slot="select-scroll-up-button"
    className={cn(
      "z-10 flex cursor-default items-center justify-center bg-popover py-1
[&_svg:not([class*='size-'])]:size-4",
      className
    )}
    {...props}
  >
    <ChevronUpIcon
    />
  </SelectPrimitive.ScrollUpButton>
 )
}

function SelectScrollDownButton({
 className,
 ...props
}: React.ComponentProps<typeof SelectPrimitive.ScrollDownButton>) {
 return (
  <SelectPrimitive.ScrollDownButton
    data-slot="select-scroll-down-button"
    className={cn(
      "z-10 flex cursor-default items-center justify-center bg-popover py-1
[&_svg:not([class*='size-'])]:size-4",
      className
    )}
    {...props}
  >
    <ChevronDownIcon
```

```
    />
   </SelectPrimitive.ScrollDownButton>
  )
}

export {
  Select,
  SelectContent,
  SelectGroup,
  SelectItem,
  SelectLabel,
  SelectScrollDownButton,
  SelectScrollUpButton,
  SelectSeparator,
  SelectTrigger,
  SelectValue,
}
```

Separator.tsx:

```
import * as React from "react"
import { Separator as SeparatorPrimitive } from "radix-ui"

import { cn } from "@/lib/utils"

function Separator({
  className,
  orientation = "horizontal",
  decorative = true,
```

```
  ...props
}: React.ComponentProps<typeof SeparatorPrimitive.Root>) {
  return (
    <SeparatorPrimitive.Root
      data-slot="separator"
      decorative={decorative}
      orientation={orientation}
      className={cn(
        "shrink-0 bg-border data-horizontal:h-px data-horizontal:w-full data-vertical:w-px
data-vertical:self-stretch",
        className
      )}
      {...props}
    />
  )
}

export { Separator }
```

sonner.tsx:

```
import { useTheme } from "next-themes"
import { Toaster as Sonner, type ToasterProps } from "sonner"
import { CircleCheckIcon, InfoIcon, TriangleAlertIcon, OctagonXIcon, Loader2Icon }
from "lucide-react"

const Toaster = ({ ...props }: ToasterProps) => {
  const { theme = "system" } = useTheme()

  return (
```

```
<Sonner
 theme={theme as ToasterProps["theme"]}
 className="toaster group"
 icons={{
  success: (
   <CircleCheckIcon className="size-4" />
  ),
  info: (
   <InfoIcon className="size-4" />
  ),
  warning: (
   <TriangleAlertIcon className="size-4" />
  ),
  error: (
   <OctagonXIcon className="size-4" />
  ),
  loading: (
   <Loader2Icon className="size-4 animate-spin" />
  ),
 }}
 style={
  {
   "--normal-bg": "var(--popover)",
   "--normal-text": "var(--popover-foreground)",
   "--normal-border": "var(--border)",
   "--border-radius": "var(--radius)",
  } as React.CSSProperties
 }
```

```
    toastOptions={{
      classNames: {
        toast: "cn-toast",
      },
    }}
    {...props}
   />
 )
}

export { Toaster }
```

tabs.tsx:

```
"use client"

import * as React from "react"
import { cva, type VariantProps } from "class-variance-authority"
import { Tabs as TabsPrimitive } from "radix-ui"

import { cn } from "@/lib/utils"

function Tabs({
  className,
  orientation = "horizontal",
  ...props
}: React.ComponentProps<typeof TabsPrimitive.Root>) {
  return (
    <TabsPrimitive.Root
```

```
    data-slot="tabs"
    data-orientation={orientation}
    className={cn(
      "group/tabs flex gap-2 data-horizontal:flex-col",
      className
    )}
    {...props}
  />
)
}

const tabsListVariants = cva(
  "group/tabs-list inline-flex w-fit items-center justify-center rounded-lg p-[3px] text-muted-foreground group-data-horizontal/tabs:h-8 group-data-vertical/tabs:h-fit group-data-vertical/tabs:flex-col data-[variant=line]:rounded-none",
  {
    variants: {
      variant: {
        default: "bg-muted",
        line: "gap-1 bg-transparent",
      },
    },
    defaultVariants: {
      variant: "default",
    },
  }
)

function TabsList({
```

```
  className,
  variant = "default",
  ...props
}: React.ComponentProps<typeof TabsPrimitive.List> &
  VariantProps<typeof tabsListVariants>) {
  return (
    <TabsPrimitive.List
      data-slot="tabs-list"
      data-variant={variant}
      className={cn(tabsListVariants({ variant }), className)}
      {...props}
    />
  )
}

function TabsTrigger({
  className,
  ...props
}: React.ComponentProps<typeof TabsPrimitive.Trigger>) {
  return (
    <TabsPrimitive.Trigger
      data-slot="tabs-trigger"
      className={cn(
        "relative inline-flex h-[calc(100%-1px)] flex-1 items-center justify-center gap-1.5
rounded-md border border-transparent px-1.5 py-0.5 text-sm font-medium whitespace-
nowrap text-foreground/60 transition-all group-data-vertical/tabs:w-full group-data-
vertical/tabs:justify-start hover:text-foreground focus-visible:border-ring focus-
visible:ring-[3px] focus-visible:ring-ring/50 focus-visible:outline-1 focus-visible:outline-
ring disabled:pointer-events-none disabled:opacity-50 dark:text-muted-foreground
dark:hover:text-foreground group-data-[variant=default]/tabs-list:data-active:shadow-
```

```
sm group-data-[variant=line]/tabs-list:data-active:shadow-none [&_svg]:pointer-events-none [&_svg]:shrink-0 [&_svg:not([class*='size-'])]:size-4",
      "group-data-[variant=line]/tabs-list:bg-transparent group-data-[variant=line]/tabs-list:data-active:bg-transparent dark:group-data-[variant=line]/tabs-list:data-active:border-transparent dark:group-data-[variant=line]/tabs-list:data-active:bg-transparent",
      "data-active:bg-background data-active:text-foreground dark:data-active:border-input dark:data-active:bg-input/30 dark:data-active:text-foreground",
      "after:absolute after:bg-foreground after:opacity-0 after:transition-opacity group-data-horizontal/tabs:after:inset-x-0 group-data-horizontal/tabs:after:bottom-[-5px] group-data-horizontal/tabs:after:h-0.5 group-data-vertical/tabs:after:inset-y-0 group-data-vertical/tabs:after:-right-1 group-data-vertical/tabs:after:w-0.5 group-data-[variant=line]/tabs-list:data-active:after:opacity-100",
      className
    )}
    {...props}
  />
 )
}

function TabsContent({
 className,
 ...props
}: React.ComponentProps<typeof TabsPrimitive.Content>) {
 return (
  <TabsPrimitive.Content
    data-slot="tabs-content"
    className={cn("flex-1 text-sm outline-none", className)}
    {...props}
  />
 )
```

```
}

export { Tabs, TabsList, TabsTrigger, TabsContent, tabsListVariants }
```

Src

Components:

CropShowcase.tsx:

```
import { useState, Suspense, useRef } from 'react';

import { Canvas, useFrame } from '@react-three/fiber';

import { useGLTF, PresentationControls, Environment, PerspectiveCamera, Float, Html }
from '@react-three/drei';

import { motion, AnimatePresence } from 'framer-motion';

import { ChevronLeft, ChevronRight, Sprout, Award, Wind, Thermometer, Droplets }
from 'lucide-react';

import { Badge } from '@/components/ui/badge';

import * as THREE from 'three';


// --- DATA ---

const cropsData = [

 {

   id: 'wheat',

   name: 'Wheat',

   scientificName: 'Triticum aestivum',

   modelPath: '/models/wheat.glb',

   color: 'from-amber-400 to-orange-600',

   accent: '#f59e0b',
```

```
  description: 'A staple grain that sustains billions. India stands as the world\'s second-largest producer, with the golden fields of Punjab and Haryana driving the nation\'s food security.',
  facts: [
    'Contributes 35% to India\'s total food grain production.',
    'Rich in B-vitamins and dietary fiber.',
    'Primary crop for the historic Green Revolution.'
  ],
  botanicalInfo: [
    { label: 'Classification', value: 'Cereal Grain', icon: <Sprout className="h-3 w-3" /> },
    { label: 'Ideal Temp', value: '12°C - 25°C', icon: <Thermometer className="h-3 w-3" /> },
    { label: 'Water Need', value: 'Moderate', icon: <Droplets className="h-3 w-3" /> }
  ],
   scale: 1,
position: [0.25, 0.1, 0] as [number, number, number],
rotation: [0, 0, 0] as [number, number, number]
 },
 {
  id: 'rice',
  name: 'Rice',
  scientificName: 'Oryza sativa',
  modelPath: '/models/rice_plant.glb',
  color: 'from-emerald-400 to-teal-600',
  accent: '#10b981',
  description: 'The lifecycle of water and sun. From the terraced hills of the Northeast to the delta plains of the South, rice is more than a crop—it\'s the heartbeat of Indian culture.',
  facts: [
```

```
    'India is the world\'s leading exporter of Basmati rice.',
    'Cultivated in over 43 million hectares across India.',
    'Requires high water retention and warm climate.'
  ],
  botanicalInfo: [
    { label: 'Classification', value: 'Grass Species', icon: <Sprout className="h-3 w-3" />
},
    { label: 'Ideal Temp', value: '25°C - 35°C', icon: <Thermometer className="h-3 w-3"
/> },
    { label: 'Water Need', value: 'High (Flood)', icon: <Droplets className="h-3 w-3" /> }
  ],
   scale: 0.06,
position: [0.2, -0.7, 0] as [number, number, number],
rotation: [0, 0, 0] as [number, number, number]
 },
 {
  id: 'corn',
  name: 'Corn',
  scientificName: 'Zea mays',
  modelPath: '/models/corn_old.glb',
  color: 'from-yellow-400 to-amber-500',
  accent: '#fbbf24',
  description: 'The versatile powerhouse. Beyond food and fodder, corn is a critical
industrial raw material, powering the biofuel revolution and global supply chains.',
  facts: [
    'Grown in nearly every Indian state across all seasons.',
    'Key ingredient for starch, oil, and ethanol production.',
    'Known as the "Queen of Cereals" due to high genetic yield.'
  ],
```

```
    botanicalInfo: [
      { label: 'Classification', value: 'Cereal Leaf', icon: <Sprout className="h-3 w-3" /> },
      { label: 'Ideal Temp', value: '18°C - 27°C', icon: <Thermometer className="h-3 w-3"
/> },
      { label: 'Water Need', value: 'Balanced', icon: <Droplets className="h-3 w-3" /> }
    ],
    scale: 0.9,
position: [0.25, -1, 0] as [number, number, number],
rotation: [0, 0, 0] as [number, number, number]
  }
];

// --- 3D COMPONENTS ---

function Model({ path, scale, position, rotation }: { path: string, scale: number, position:
[number, number, number], rotation: [number, number, number] }) {
  const { scene } = useGLTF(path);
  const meshRef = useRef<any>(null);

  useFrame((state) => {
    if (meshRef.current) {
      meshRef.current.rotation.y += 0.005;
      meshRef.current.position.y = position[1] + Math.sin(state.clock.getElapsedTime()) *
0.1;
    }
  });

  return (
    <primitive
```

```
    ref={meshRef}
    object={scene}
    scale={scale}
    position={position}
    rotation={rotation}
   />
  );
}

const CropInfoCard = ({ crop, isVisible }: { crop: typeof cropsData[0], isVisible: boolean
}) => {
  return (
   <motion.div
    initial={{ opacity: 0, x: 50 }}
    animate={{ opacity: isVisible ? 1 : 0, x: isVisible ? 0 : 50 }}
    transition={{ duration: 0.6, ease: "easeOut" }}
    className="absolute right-4 md:right-12 top-1/2 -translate-y-1/2 w-full max-w-
[320px] md:max-w-md z-20"
   >
    <div className="backdrop-blur-xl bg-white/10 dark:bg-slate-900/40 border border-
white/20 dark:border-white/10 rounded-[2rem] p-6 md:p-8 shadow-2xl overflow-hidden
relative group">
     {/* Ambient Glow */}
     <div className={`absolute -top-24 -right-24 w-48 h-48 bg-gradient-to-br
${crop.color} opacity-20 blur-3xl rounded-full`} />


     <div className="relative z-10">
      <div className="flex items-center gap-3 mb-4">
       <Badge variant="outline" className="bg-white/5 border-white/20 text-white/80
px-3 py-1 rounded-full text-[10px] tracking-widest uppercase font-bold">
```

```
          {crop.scientificName}
         </Badge>
         <div className={` h-2 w-2 rounded-full animate-pulse bg-gradient-to-r ${crop.color}` } />
        </div>

        <h2 className="text-4xl md:text-5xl font-black text-white mb-4 tracking-tighter">
          {crop.name}
        </h2>

        <p className="text-white/70 text-sm md:text-base leading-relaxed mb-8 font-medium">
          {crop.description}
        </p>

        <div className="space-y-6">
          <h4 className="text-xs font-bold text-white/40 uppercase tracking-[0.2em] flex items-center gap-2">
            <Wind className="h-3 w-3" /> Botanical Profile
          </h4>

          <div className="space-y-4">
            {crop.botanicalInfo.map((info, i) => (
              <div key={i} className="flex items-center justify-between p-3 rounded-xl bg-white/5 border border-white/10 group/item hover:bg-white/10 transition-colors">
                <div className="flex items-center gap-3">
                  <div className={` p-1.5 rounded-lg bg-gradient-to-br ${crop.color} text-white` }>
                    {info.icon}
```

```
              </div>
              <span className="text-white/60 font-medium text-xs uppercase tracking-wider">{info.label}</span>
            </div>
            <span className="text-white font-bold text-sm">
              {info.value}
            </span>
          </div>
        ))}
      </div>
    </div>

    <div className="mt-8 pt-6 border-t border-white/10 flex items-center justify-between">
      <span className="text-white/30 text-[10px] font-bold uppercase tracking-[0.2em]">Scientific Data 2026</span>
      <div className="flex gap-1">
        <div className={` h-1 w-4 rounded-full bg-gradient-to-r ${crop.color}` } />
        <div className="h-1 w-1 rounded-full bg-white/20" />
        <div className="h-1 w-1 rounded-full bg-white/20" />
      </div>
    </div>
  </div>
 </div>
</motion.div>
);
};

export default function CropShowcase() {
```

```
  const [currentIndex, setCurrentIndex] = useState(0);
  const touchStartX = useRef<number | null>(null);

  const nextCrop = () => setCurrentIndex((prev) => (prev + 1) % cropsData.length);
  const prevCrop = () => setCurrentIndex((prev) => (prev - 1 + cropsData.length) %
cropsData.length);

  const handleTouchStart = (e: React.TouchEvent) => {
    touchStartX.current = e.touches[0].clientX;
  };

  const handleTouchEnd = (e: React.TouchEvent) => {
    if (touchStartX.current === null) return;
    const touchEndX = e.changedTouches[0].clientX;
    const diff = touchStartX.current - touchEndX;

    if (Math.abs(diff) > 50) {
      if (diff > 0) nextCrop();
      else prevCrop();
    }
    touchStartX.current = null;
  };

  const activeCrop = cropsData[currentIndex];

  return (
    <section
      id="crops"
```

```
    className="relative w-screen h-screen overflow-hidden bg-[#0a0a0a]"
    onTouchStart={handleTouchStart}
    onTouchEnd={handleTouchEnd}
  >
    {/* Background elements */}
    <div className="absolute inset-0 z-0">
      <div className={` absolute top-1/2 left-1/4 -translate-y-1/2 w-[800px] h-[800px] bg-gradient-to-br ${activeCrop.color} opacity-[0.03] blur-[150px] rounded-full transition-all duration-1000 `} />
      <div className="absolute inset-0 bg-[radial-gradient(circle_at_20%_30%,rgba(255,255,255,0.02)_0%,transparent_50%)]" />
    </div>

    {/* Navigation Layer */}
    <div className="absolute inset-0 z-30 pointer-events-none flex items-center justify-between px-6 md:px-12">
      <button
        onClick={prevCrop}
        className="pointer-events-auto h-12 w-12 md:h-14 md:w-14 rounded-full bg-white/5 border border-white/10 flex items-center justify-center text-white hover:bg-white/10 hover:border-white/20 transition-all group active:scale-90"
      >
        <ChevronLeft className="h-6 w-6 group-hover:-translate-x-1 transition-transform" />
      </button>
      <button
        onClick={nextCrop}
        className="pointer-events-auto h-12 w-12 md:h-14 md:w-14 rounded-full bg-white/5 border border-white/10 flex items-center justify-center text-white hover:bg-white/10 hover:border-white/20 transition-all group active:scale-90"
      >
```

```
        <ChevronRight className="h-6 w-6 group-hover:translate-x-1 transition-transform" />
      </button>
    </div>

    {/* Hero Stats (Top Left) */}
    <div className="absolute top-32 left-8 md:left-12 z-20 pointer-events-none">
      <motion.div
        key={activeCrop.id}
        initial={{ opacity: 0, y: 20 }}
        animate={{ opacity: 1, y: 0 }}
        className="flex flex-col gap-8"
      >
        <div className="flex flex-col gap-1">
          <span className="text-[10px] font-black text-white/30 uppercase tracking-[0.4em]">Agricultural Insights</span>
          <div className="h-1 w-12 bg-white/20 mt-2 rounded-full overflow-hidden">
            <motion.div
              initial={{ x: '-100%' }}
              animate={{ x: '0%' }}
              transition={{ duration: 0.8 }}
              className={` h-full w-full bg-gradient-to-r ${activeCrop.color}` }
            />
          </div>
        </div>

        <div className="space-y-6">
          {activeCrop.facts.map((fact, i) => (
            <motion.div
```

```
            key={i}
            initial={{ opacity: 0, x: -20 }}
            animate={{ opacity: 1, x: 0 }}
            transition={{ delay: 0.2 + i * 0.1 }}
            className="flex items-start gap-4 max-w-[280px]"
          >
            <div className="mt-1 p-1 rounded-md bg-white/5 border border-white/10 shrink-0">
              <Award className={` h-3 w-3 text-transparent bg-clip-text bg-gradient-to-r ${activeCrop.color}` } />
            </div>
            <p className="text-white/60 text-xs font-semibold leading-relaxed uppercase tracking-wider">
              {fact}
            </p>
          </motion.div>
        ))}
      </div>
    </motion.div>
  </div>

  {/* Progress Indicator */}
  <div className="absolute bottom-12 left-1/2 -translate-x-1/2 z-20 flex gap-4">
    {cropsData.map((_, i) => (
      <button
        key={i}
        onClick={() => setCurrentIndex(i)}
        className={` h-1.5 transition-all duration-500 rounded-full ${i === currentIndex ? 'w-12 bg-white' : 'w-4 bg-white/20 hover:bg-white/40'}` }
```

```
      />
    ))}
  </div>

  {/* 3D Canvas */}
  <div className="absolute inset-0 z-10">
    <Canvas
      shadows={{ type: THREE.PCFShadowMap }}
      gl={{
        antialias: true,
        alpha: true,
        precision: 'highp'
      }}
    >
      <PerspectiveCamera makeDefault position={[0, 0, 8]} fov={35} />
      <ambientLight intensity={0.5} />
      <spotLight position={[10, 10, 10]} angle={0.15} penumbra={1} intensity={2}
castShadow />
      <pointLight position={[-10, -10, -10]} intensity={1} />

      <Suspense fallback={<Html center><div className="text-white font-black text-2xl
tracking-tighter animate-pulse uppercase">Scaling...</div></Html>}>
        <Float
          speed={2}
          rotationIntensity={0.5}
          floatIntensity={0.5}
        >
        <PresentationControls
          global
```

```
            snap
            rotation={[0, 0.3, 0]}
            polar={[-Math.PI / 3, Math.PI / 3]}
            azimuth={[-Math.PI / 1.4, Math.PI / 1.4]}
           >
            <group scale={1.8} position={[-1.2, 0, 0]}>
              <Model
                key={activeCrop.id}
                path={activeCrop.modelPath}
                scale={activeCrop.scale}
                position={activeCrop.position}
                rotation={activeCrop.rotation}
              />
            </group>
           </PresentationControls>
          </Float>
          <Environment preset="forest" />
        </Suspense>
      </Canvas>
    </div>

    {/* Info Card Overlay */}
    <AnimatePresence mode="wait">
      <CropInfoCard key={activeCrop.id} crop={activeCrop} isVisible={true} />
    </AnimatePresence>
  </section>
 );
}
```

DarkVeil.tsx:

```
import { useRef, useEffect } from 'react';
import { Renderer, Program, Mesh, Triangle, Vec2 } from 'ogl';

const vertex = `
attribute vec2 position;
void main(){gl_Position=vec4(position,0.0,1.0);}
`;

const fragment = `
#ifdef GL_ES
precision lowp float;
#endif
uniform vec2 uResolution;
uniform float uTime;
uniform float uHueShift;
uniform float uNoise;
uniform float uScan;
uniform float uScanFreq;
uniform float uWarp;
#define iTime uTime
#define iResolution uResolution

vec4 buf[8];
float rand(vec2 c){return fract(sin(dot(c,vec2(12.9898,78.233)))*43758.5453);}

mat3 rgb2yiq=mat3(0.299,0.587,0.114,0.596,-0.274,-0.322,0.211,-0.523,0.312);
```

```
mat3 yiq2rgb=mat3(1.0,0.956,0.621,1.0,-0.272,-0.647,1.0,-1.106,1.703);

vec3 hueShiftRGB(vec3 col,float deg){
   vec3 yiq=rgb2yiq*col;
   float rad=radians(deg);
   float cosh=cos(rad),sinh=sin(rad);
   vec3 yiqShift=vec3(yiq.x,yiq.y*cosh-yiq.z*sinh,yiq.y*sinh+yiq.z*cosh);
   return clamp(yiq2rgb*yiqShift,0.0,1.0);
}

vec4 sigmoid(vec4 x){return 1./(1.+exp(-x));}

vec4 cppn_fn(vec2 coordinate,float in0,float in1,float in2){
   buf[6]=vec4(coordinate.x,coordinate.y,0.3948333106474662+in0,0.36+in1);
   buf[7]=vec4(0.14+in2,sqrt(coordinate.x*coordinate.x+coordinate.y*coordinate.y),0.,0.
);
   buf[0]=mat4(vec4(6.5404263,-3.6126034,0.7590882,-
1.13613),vec4(2.4582713,3.1660357,1.2219609,0.06276096),vec4(-5.478085,-
6.159632,1.8701609,-4.7742867),vec4(6.039214,-5.542865,-
0.90925294,3.251348))*buf[6]+mat4(vec4(0.8473259,-
5.722911,3.975766,1.6522468),vec4(-0.24321538,0.5839259,-1.7661959,-
5.350116),vec4(0.,0.,0.,0.),vec4(0.,0.,0.,0.))*buf[7]+vec4(0.21808943,1.1243913,-
1.7969975,5.0294676);
   buf[1]=mat4(vec4(-3.3522482,-6.0612736,0.55641043,-
4.4719114),vec4(0.8631464,1.7432913,5.643898,1.6106541),vec4(2.4941394,-
3.5012043,1.7184316,6.357333),vec4(3.310376,8.209261,1.1355612,-
1.165539))*buf[6]+mat4(vec4(5.24046,-
13.034365,0.009859298,15.870829),vec4(2.987511,3.129433,-0.89023495,-
1.6822904),vec4(0.,0.,0.,0.),vec4(0.,0.,0.,0.))*buf[7]+vec4(-5.9457836,-6.573602,-
0.8812491,1.5436668);
   buf[0]=sigmoid(buf[0]);buf[1]=sigmoid(buf[1]);
```

```
  buf[2]=mat4(vec4(-15.219568,8.095543,-2.429353,-1.9381982),vec4(-
5.951362,4.3115187,2.6393783,1.274315),vec4(-
7.3145227,6.7297835,5.2473326,5.9411426),vec4(5.0796127,8.979051,-1.7278991,-
1.158976))*buf[6]+mat4(vec4(-11.967154,-
11.608155,6.1486754,11.237008),vec4(2.124141,-6.263192,-1.7050359,-
0.7021966),vec4(0.,0.,0.,0.),vec4(0.,0.,0.,0.))*buf[7]+vec4(-4.17164,-3.2281182,-
4.576417,-3.6401186);

  buf[3]=mat4(vec4(3.1832156,-
13.738922,1.879223,3.233465),vec4(0.64300746,12.768129,1.9141049,0.50990224),v
ec4(-
0.049295485,4.4807224,1.4733979,1.801449),vec4(5.0039253,13.000481,3.3991797,-
4.5561905))*buf[6]+mat4(vec4(-0.1285731,7.720628,-
3.1425676,4.742367),vec4(0.6393625,3.714393,-0.8108378,-
0.39174938),vec4(0.,0.,0.,0.),vec4(0.,0.,0.,0.))*buf[7]+vec4(-1.1811101,-
21.621881,0.7851888,1.2329718);

  buf[2]=sigmoid(buf[2]);buf[3]=sigmoid(buf[3]);

  buf[4]=mat4(vec4(5.214916,-7.183024,2.7228765,2.6592617),vec4(-5.601878,-
25.3591,4.067988,0.4602802),vec4(-
10.57759,24.286327,21.102104,37.546658),vec4(4.3024497,-1.9625226,2.3458803,-
1.372816))*buf[0]+mat4(vec4(-17.6526,-
10.507558,2.2587414,12.462782),vec4(6.265566,-502.75443,-
12.642513,0.9112289),vec4(-10.983244,20.741234,-9.701768,-
0.7635988),vec4(5.383626,1.4819539,-4.1911616,-
4.8444734))*buf[1]+mat4(vec4(12.785233,-16.345072,-0.39901125,1.7955981),vec4(-
30.48365,-1.8345358,1.4542528,-1.1118771),vec4(19.872723,-7.337935,-42.941723,-
98.52709),vec4(8.337645,-2.7312303,-2.2927687,-36.142323))*buf[2]+mat4(vec4(-
16.298317,3.5471997,-0.44300047,-9.444417),vec4(57.5077,-35.609753,16.163465,-
4.1534753),vec4(-0.07470326,-3.8656476,-7.0901804,3.1523974),vec4(-12.559385,-
7.077619,1.490437,-0.8211543))*buf[3]+vec4(-7.67914,15.927437,1.3207729,-
1.6686112);

  buf[5]=mat4(vec4(-1.4109162,-0.372762,-3.770383,-21.367174),vec4(-6.2103205,-
9.35908,0.92529047,8.82561),vec4(11.460242,-22.348068,13.625772,-
18.693201),vec4(-0.3429052,-3.9905605,-2.4626114,-
0.45033523))*buf[0]+mat4(vec4(7.3481627,-4.3661838,-6.3037653,-
3.868115),vec4(1.5462853,6.5488915,1.9701879,-0.58291394),vec4(6.5858274,-
2.2180402,3.7127688,-1.3730392),vec4(-5.7973905,10.134961,-2.3395722,-
5.965605))*buf[1]+mat4(vec4(-2.5132585,-6.6685553,-1.4029363,-0.16285264),vec4(-
0.37908727,0.53738135,4.389061,-1.3024765),vec4(-0.70647055,2.0111287,-
5.1659346,-3.728635),vec4(-13.562562,10.487719,-0.9173751,-
```

2.6487076))*buf[2]+mat4(vec4(-8.645013,6.5546675,-6.3944063,-5.5933375),vec4(-0.57783127,-1.077275,36.91025,5.736769),vec4(14.283112,3.7146652,7.1452246,-4.5958776),vec4(2.7192075,3.6021907,-4.366337,-2.3653464))*buf[3]+vec4(-5.9000807,-4.329569,1.2427121,8.59503);

buf[4]=sigmoid(buf[4]);buf[5]=sigmoid(buf[5]);

buf[6]=mat4(vec4(-1.61102,0.7970257,1.4675229,0.20917463),vec4(-28.793737,-7.1390953,1.5025433,4.656581),vec4(-10.94861,39.66238,0.74318546,-10.095605),vec4(-0.7229728,-1.5483948,0.7301322,2.1687684))*buf[0]+mat4(vec4(3.2547753,21.489103,-1.0194173,-3.3100595),vec4(-3.7316632,-3.3792162,-7.223193,-0.23685838),vec4(13.1804495,0.7916005,5.338587,5.687114),vec4(-4.167605,-17.798311,-6.815736,-1.6451967))*buf[1]+mat4(vec4(0.604885,-7.800309,-7.213122,-2.741014),vec4(-3.522382,-0.12359311,-0.5258442,0.43852118),vec4(9.6752825,-22.853785,2.062431,0.099892326),vec4(-4.3196306,-17.730087,2.5184598,5.30267))*buf[2]+mat4(vec4(-6.545563,-15.790176,-6.0438633,-5.415399),vec4(-43.591583,28.551912,-16.00161,18.84728),vec4(4.212382,8.394307,3.0958717,8.657522),vec4(-5.0237565,-4.450633,-4.4768,-5.5010443))*buf[3]+mat4(vec4(1.6985557,-67.05806,6.897715,1.9004834),vec4(1.8680354,2.3915145,2.5231109,4.081538),vec4(11.158006,1.7294737,2.0738268,7.386411),vec4(-4.256034,-306.24686,8.258898,-17.132736))*buf[4]+mat4(vec4(1.6889864,-4.5852966,3.8534803,-6.3482175),vec4(1.3543309,-1.2640043,9.932754,2.9079645),vec4(-5.2770967,0.07150358,-0.13962056,3.3269649),vec4(28.34703,-4.918278,6.1044083,4.085355))*buf[5]+vec4(6.6818056,12.522166,-3.7075126,-4.104386);

buf[7]=mat4(vec4(-8.265602,-4.7027016,5.098234,0.7509808),vec4(8.6507845,-17.15949,16.51939,-8.884479),vec4(-4.036479,-2.3946867,-2.6055532,-1.9866527),vec4(-2.2167742,-1.8135649,-5.9759874,4.8846445))*buf[0]+mat4(vec4(6.7790847,3.5076547,-2.8191125,-2.7028968),vec4(-5.743024,-0.27844876,1.4958696,-5.0517144),vec4(13.122226,15.735168,-2.9397483,-4.101023),vec4(-14.375265,-5.030483,-6.2599335,2.9848232))*buf[1]+mat4(vec4(4.0950394,-0.94011575,-5.674733,4.755022),vec4(4.3809423,4.8310084,1.7425908,-3.437416),vec4(2.117492,0.16342592,-104.56341,16.949184),vec4(-5.22543,-2.994248,3.8350096,-1.9364246))*buf[2]+mat4(vec4(-5.900337,1.7946124,-13.604192,-3.8060522),vec4(6.6583457,31.911177,25.164474,91.81147),vec4(11.840538,4.1503043,-0.7314397,6.768467),vec4(-6.3967767,4.034772,6.1714606,-0.32874924))*buf[3]+mat4(vec4(3.4992442,-196.91893,-

```
8.923708,2.8142626),vec4(3.4806502,-3.1846354,5.1725626,5.1804223),vec4(-
2.4009497,15.585794,1.2863957,2.0252278),vec4(-71.25271,-62.441242,-
8.138444,0.50670296))*buf[4]+mat4(vec4(-12.291733,-11.176166,-
7.3474145,4.390294),vec4(10.805477,5.6337385,-0.9385842,-4.7348723),vec4(-
12.869276,-
7.039391,5.3029537,7.5436664),vec4(1.4593618,8.91898,3.5101583,5.840625))*buf[5
]+vec4(2.2415268,-6.705987,-0.98861027,-2.117676);

  buf[6]=sigmoid(buf[6]);buf[7]=sigmoid(buf[7]);

  buf[0]=mat4(vec4(1.6794263,1.3817469,2.9625452,0.),vec4(-1.8834411,-1.4806935,-
3.5924516,0.),vec4(-1.3279216,-1.0918057,-
2.3124623,0.),vec4(0.2662234,0.23235129,0.44178495,0.))*buf[0]+mat4(vec4(-
0.6299101,-0.5945583,-
0.9125601,0.),vec4(0.17828953,0.18300213,0.18182953,0.),vec4(-2.96544,-
2.5819945,-
4.9001055,0.),vec4(1.4195864,1.1868085,2.5176322,0.))*buf[1]+mat4(vec4(-
1.2584374,-1.0552157,-2.1688404,0.),vec4(-0.7200217,-0.52666044,-
1.438251,0.),vec4(0.15345335,0.15196142,0.272854,0.),vec4(0.945728,0.8861938,1.2
766753,0.))*buf[2]+mat4(vec4(-2.4218085,-1.968602,-4.35166,0.),vec4(-22.683098,-
18.0544,-41.954372,0.),vec4(0.63792,0.5470648,1.1078634,0.),vec4(-1.5489894,-
1.3075932,-2.6444845,0.))*buf[3]+mat4(vec4(-0.49252132,-0.39877754,-
0.91366625,0.),vec4(0.95609266,0.7923952,1.640221,0.),vec4(0.30616966,0.1569392
5,0.8639857,0.),vec4(1.1825981,0.94504964,2.176963,0.))*buf[4]+mat4(vec4(0.35446
745,0.3293795,0.59547555,0.),vec4(-0.58784515,-0.48177817,-
1.0614829,0.),vec4(2.5271258,1.9991658,4.6846647,0.),vec4(0.13042648,0.08864098,
0.30187556,0.))*buf[5]+mat4(vec4(-1.7718065,-1.4033192,-
3.3355875,0.),vec4(3.1664357,2.638297,5.378702,0.),vec4(-3.1724713,-2.6107926,-
5.549295,0.),vec4(-2.851368,-2.249092,-
5.3013067,0.))*buf[6]+mat4(vec4(1.5203838,1.2212278,2.8404984,0.),vec4(1.5210563
,1.2651345,2.683903,0.),vec4(2.9789467,2.4364579,5.2347264,0.),vec4(2.2270417,1.8
825914,3.8028636,0.))*buf[7]+vec4(-1.5468478,-3.6171484,0.24762098,0.);

  buf[0]=sigmoid(buf[0]);

  return vec4(buf[0].x,buf[0].y,buf[0].z,1.);

}


void mainImage(out vec4 fragColor,in vec2 fragCoord){

  vec2 uv=fragCoord/uResolution.xy*2.-1.;

  uv.y*=-1.;
```

```
  uv+=uWarp*vec2(sin(uv.y*6.283+uTime*0.5),cos(uv.x*6.283+uTime*0.5))*0.05;
  fragColor=cppn_fn(uv,0.1*sin(0.3*uTime),0.1*sin(0.69*uTime),0.1*sin(0.44*uTime));
}

void main(){
  vec4 col;mainImage(col,gl_FragCoord.xy);
  col.rgb=hueShiftRGB(col.rgb,uHueShift);
  float scanline_val=sin(gl_FragCoord.y*uScanFreq)*0.5+0.5;
  col.rgb*=1.-(scanline_val*scanline_val)*uScan;
  col.rgb+=(rand(gl_FragCoord.xy+uTime)-0.5)*uNoise;
  gl_FragColor=vec4(clamp(col.rgb,0.0,1.0),1.0);
}
`;

type Props = {
 hueShift?: number;
 noiseIntensity?: number;
 scanlineIntensity?: number;
 speed?: number;
 scanlineFrequency?: number;
 warpAmount?: number;
 resolutionScale?: number;
};

export default function DarkVeil({
 hueShift = 0,
 noiseIntensity = 0,
 scanlineIntensity = 0,
```

```
  speed = 0.5,
  scanlineFrequency = 0,
  warpAmount = 0,
  resolutionScale = 1
}: Props) {
  const ref = useRef<HTMLCanvasElement>(null);
  useEffect(() => {
    const canvas = ref.current as HTMLCanvasElement;
    const parent = canvas.parentElement as HTMLElement;

    const renderer = new Renderer({
      dpr: Math.min(window.devicePixelRatio, 2),
      canvas
    });

    const gl = renderer.gl;
    const geometry = new Triangle(gl);

    const program = new Program(gl, {
      vertex,
      fragment,
      uniforms: {
        uTime: { value: 0 },
        uResolution: { value: new Vec2() },
        uHueShift: { value: hueShift },
        uNoise: { value: noiseIntensity },
        uScan: { value: scanlineIntensity },
        uScanFreq: { value: scanlineFrequency },
```

```
    uWarp: { value: warpAmount }
  }
});

const mesh = new Mesh(gl, { geometry, program });

const resize = () => {
  const w = parent.clientWidth,
    h = parent.clientHeight;
  renderer.setSize(w * resolutionScale, h * resolutionScale);
  program.uniforms.uResolution.value.set(w, h);
};

window.addEventListener('resize', resize);
resize();

const start = performance.now();
let frame = 0;

const loop = () => {
  program.uniforms.uTime.value = ((performance.now() - start) / 1000) * speed;
  program.uniforms.uHueShift.value = hueShift;
  program.uniforms.uNoise.value = noiseIntensity;
  program.uniforms.uScan.value = scanlineIntensity;
  program.uniforms.uScanFreq.value = scanlineFrequency;
  program.uniforms.uWarp.value = warpAmount;
  renderer.render({ scene: mesh });
  frame = requestAnimationFrame(loop);
```

```
    };

    loop();

    return () => {
      cancelAnimationFrame(frame);
      window.removeEventListener('resize', resize);
    };
  }, [hueShift, noiseIntensity, scanlineIntensity, speed, scanlineFrequency, warpAmount, resolutionScale]);
  return <canvas ref={ref} className="w-full h-full block" />;
}
```

LanguageSwitcher.tsx:

```
import { useTranslationText } from "@/hooks/useTranslationText";
import {
    Select,
    SelectContent,
    SelectItem,
    SelectTrigger,
    SelectValue,
} from "@/components/ui/select";
import { Globe } from "lucide-react";
import axios from "axios";

export function LanguageSwitcher() {
    const { i18n, t } = useTranslationText();
```

```
const handleLanguageChange = async (lng: string) => {
  await i18n.changeLanguage(lng);

  // Attempt to update backend if a token exists
  const token = localStorage.getItem("token");
  if (token) {
    try {
      await axios.put(
        "http://localhost:8000/api/farmers/me",
        { preferred_language: lng },
        { headers: { Authorization: `Bearer ${token}` } }
      );
    } catch (error) {
      console.error("Failed to update language on backend", error);
    }
  }
};

return (
  <Select value={i18n.language || "en"} onValueChange={handleLanguageChange}>
    <SelectTrigger className="w-fit min-w-[130px]">
      <Globe className="mr-2 h-4 w-4" />
      <SelectValue placeholder={t('common.language')} />
    </SelectTrigger>
    <SelectContent>
      <SelectItem value="en">English</SelectItem>
      <SelectItem value="hi">हिंदी</SelectItem>
      <SelectItem value="mr">मराठी</SelectItem>
```

```
            <SelectItem value="gu">ગુજરાતી</SelectItem>
          </SelectContent>
        </Select>
      );
}
```

Navbar.tsx:

```
import React, { useState, useEffect } from 'react';
import { Link, useLocation } from 'react-router-dom';
import { motion } from 'framer-motion';
import { Button } from '@/components/ui/button';
import { Leaf, Sun, Moon } from 'lucide-react';
import { useTheme } from '@/components/theme-provider';
import { useTranslationText } from '@/hooks/useTranslationText';
import { LanguageSwitcher } from '@/components/LanguageSwitcher';

export function Navbar() {
  const { theme, setTheme } = useTheme();
  const { t } = useTranslationText();
  const location = useLocation();
  const [isLoggedIn, setIsLoggedIn] = useState(!!localStorage.getItem('access_token'));

  useEffect(() => {
    const checkAuth = () => setIsLoggedIn(!!localStorage.getItem('access_token'));
    checkAuth(); // Initial check on mount/location change

    // Listen for storage changes (works across tabs)
    window.addEventListener('storage', checkAuth);
```

```
    return () => window.removeEventListener('storage', checkAuth);
  }, [location]);


  return (
    <nav className="fixed top-0 w-full z-50 px-4 pt-4 pointer-events-none">
      <motion.div
        initial={{ y: -24, opacity: 0 }}
        animate={{ y: 0, opacity: 1 }}
        transition={{ duration: 0.6, ease: "easeOut" }}
        className="max-w-7xl mx-auto flex items-center justify-between px-5 py-3 bg-white/80 dark:bg-[#060a0f]/80 backdrop-blur-md rounded-2xl border border-slate-200 dark:border-slate-800 pointer-events-auto shadow-sm"
      >
        <Link to="/" className="flex items-center gap-2.5 group cursor-pointer">
          <div className="bg-gradient-to-br from-emerald-500 to-emerald-700 p-1.5 rounded-xl group-hover:scale-110 transition-transform shadow-lg shadow-emerald-500/25">
            <Leaf className="h-5 w-5 text-white" />
          </div>
          <span className="text-lg font-black tracking-[-0.04em] text-emerald-900 dark:text-white">
            AGRISENSE
          </span>
        </Link>


        <div className="hidden md:flex gap-7 text-sm font-medium text-slate-600 dark:text-slate-400">
          <Link to="/#features" className="hover:text-emerald-600 dark:hover:text-emerald-400 transition-colors">
            {t('nav.features') || 'Features'}
```

```
            </Link>
            <Link to="/#process" className="hover:text-emerald-600 dark:hover:text-emerald-400 transition-colors">
              {t('nav.process') || 'Process'}
            </Link>
            <Link to="/#testimonials" className="hover:text-emerald-600 dark:hover:text-emerald-400 transition-colors">
              {t('nav.testimonials') || 'Testimonials'}
            </Link>
            <Link to="/#faq" className="hover:text-emerald-600 dark:hover:text-emerald-400 transition-colors">
              {t('nav.faq') || 'FAQ'}
            </Link>
          </div>

          <div className="flex items-center gap-2">
            <Button
              variant="ghost"
              size="icon"
              onClick={() => setTheme(theme === "dark" ? "light" : "dark")}
              className="h-9 w-9 text-slate-600 dark:text-slate-400 hover:text-emerald-600 dark:hover:text-emerald-400 hover:bg-emerald-50 dark:hover:bg-emerald-950 rounded-xl"
            >
              {theme === "dark" ? <Sun className="h-4 w-4" /> : <Moon className="h-4 w-4" />}
            </Button>
            <LanguageSwitcher />

            {isLoggedIn ? (
```

```
                <Link to="/dashboard">
                  <Button className="h-9 px-5 text-sm bg-emerald-600 hover:bg-emerald-500 text-white font-semibold rounded-xl shadow-lg shadow-emerald-500/25 transition-all hover:scale-105 hover:shadow-emerald-500/40">
                    {t('dashboard.title') || 'Dashboard'}
                  </Button>
                </Link>
              ) : (
                <>
                  <Link to="/auth">
                    <Button variant="ghost" className="h-9 px-4 text-sm font-semibold text-slate-700 dark:text-slate-300 hover:text-emerald-700 dark:hover:text-white hover:bg-emerald-50 dark:hover:bg-white/5 rounded-xl">
                      {t('nav.login') || 'Login'}
                    </Button>
                  </Link>
                  <Link to="/auth">
                    <Button className="h-9 px-5 text-sm bg-emerald-600 hover:bg-emerald-500 text-white font-semibold rounded-xl shadow-lg shadow-emerald-500/25 transition-all hover:scale-105 hover:shadow-emerald-500/40">
                      {t('nav.get_started') || 'Get Started'}
                    </Button>
                  </Link>
                </>
              )}
            </div>
          </motion.div>
        </nav>
      );
    }
```

theme-provider.tsx:

```
import React, { createContext, useContext, useEffect, useState } from "react"

type Theme = "dark" | "light" | "system"

type ThemeProviderProps = {
  children: React.ReactNode
  defaultTheme?: Theme
  storageKey?: string
}

type ThemeProviderState = {
  theme: Theme
  setTheme: (theme: Theme) => void
}

const initialState: ThemeProviderState = {
  theme: "system",
  setTheme: () => null,
}

const ThemeProviderContext = createContext<ThemeProviderState>(initialState)

export function ThemeProvider({
  children,
  defaultTheme = "dark", // Setting Dark Mode as Default!
  storageKey = "vite-ui-theme",
```

```
  ...props
}: ThemeProviderProps) {
  const [theme, setTheme] = useState<Theme>(
    () => (localStorage.getItem(storageKey) as Theme) || defaultTheme
  )

  useEffect(() => {
    const root = window.document.documentElement

    root.classList.remove("light", "dark")

    if (theme === "system") {
      const systemTheme = window.matchMedia("(prefers-color-scheme: dark)")
        .matches
        ? "dark"
        : "light"

      root.classList.add(systemTheme)
      return
    }

    root.classList.add(theme)
  }, [theme])

  const value = {
    theme,
    setTheme: (theme: Theme) => {
      localStorage.setItem(storageKey, theme)
```

```
    setTheme(theme)
  },
 }

 return (
  <ThemeProviderContext.Provider {...props} value={value}>
    {children}
  </ThemeProviderContext.Provider>
 )
}

export const useTheme = () => {
 const context = useContext(ThemeProviderContext)
 if (context === undefined)
  throw new Error("useTheme must be used within a ThemeProvider")
 return context
}
```

Src

Pages:

App.css:

```
#root {
  max-width: 1280px;
  margin: 0 auto;
  padding: 2rem;
  text-align: center;
}

.logo {
  height: 6em;
  padding: 1.5em;
  will-change: filter;
  transition: filter 300ms;
}
.logo:hover {
  filter: drop-shadow(0 0 2em #646cffaa);
}
.logo.react:hover {
  filter: drop-shadow(0 0 2em #61dafbaa);
}

@keyframes logo-spin {
  from {
    transform: rotate(0deg);
  }
  to {
```

```
    transform: rotate(360deg);
  }
}

@media (prefers-reduced-motion: no-preference) {
  a:nth-of-type(2) .logo {
    animation: logo-spin infinite 20s linear;
  }
}

.card {
  padding: 2em;
}

.read-the-docs {
  color: #888;
}
```

App.tsx:

```
// App.tsx
import { BrowserRouter as Router, Routes, Route, Navigate } from 'react-router-dom';
import LandingPage from './pages/LandingPage';
import AuthPage from './pages/AuthPage';
import ProfileWizard from './pages/ProfileWizard';
import Dashboard from './pages/Dashboard';
import { Toaster } from "@/components/ui/sonner";
import { ThemeProvider } from "@/components/theme-provider";
import { Navbar } from "@/components/Navbar";
```

```
function App() {
  return (
    // Wrap the entire app in the ThemeProvider
    <ThemeProvider defaultTheme="dark" storageKey="agrisense-theme">
      <Router>
        {/* Sonner Toaster must be inside the Router provider */}
        <Toaster position="top-center" richColors closeButton />
        <Navbar />

        <Routes>
          {/* Public Routes */}
          <Route path="/" element={<LandingPage />} />
          <Route path="/auth" element={<AuthPage />} />

          {/* Farmer Workflow Routes */}
          <Route path="/profile-setup" element={<ProfileWizard />} />
          <Route path="/dashboard" element={<Dashboard />} />

          {/* Catch-all: Redirect to Landing */}
          <Route path="*" element={<Navigate to="/" replace />} />
        </Routes>
      </Router>
    </ThemeProvider>
  );
}

export default App;
```

index.css:

```
@tailwind base;
@tailwind components;
@tailwind utilities;

@layer base {
  :root {
    --background: 0 0% 100%;
    --foreground: 222.2 84% 4.9%;
    --card: 0 0% 100%;
    --card-foreground: 222.2 84% 4.9%;
    --popover: 0 0% 100%;
    --popover-foreground: 222.2 84% 4.9%;
    --primary: 142.1 76.2% 36.3%;
    --primary-foreground: 355.7 100% 97.3%;
    --secondary: 210 40% 96.1%;
    --secondary-foreground: 222.2 47.4% 11.2%;
    --muted: 210 40% 96.1%;
    --muted-foreground: 215.4 16.3% 46.9%;
    --accent: 210 40% 96.1%;
    --accent-foreground: 222.2 47.4% 11.2%;
    --destructive: 0 84.2% 60.2%;
    --destructive-foreground: 210 40% 98%;
    --border: 214.3 31.8% 91.4%;
    --input: 214.3 31.8% 91.4%;
    --ring: 142.1 76.2% 36.3%;
    --radius: 0.5rem;
  }
```

```
}

@layer base {
  * {
    border-color: hsl(var(--border));
  }
  body {
    @apply bg-background text-foreground;
  }
}
```

Main.jsx:

```
import React, { Suspense } from "react";
import ReactDOM from "react-dom/client";
import App from "./App";
import "./index.css";
import "./i18n/i18n";

ReactDOM.createRoot(document.getElementById("root")).render(
  <React.StrictMode>
    <Suspense fallback={<div>Loading translation...</div>}>
      <App />
    </Suspense>
  </React.StrictMode>,
);
```